ActInf GuestStream #013.1 ~ Adam Safron

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hi i'm blue knight i'm here with daniel friedman today is november 8th 2021 for a guest stream with adam saffron adam please take it away um hi blue hi daniel hi everyone it's good to be talking with you today and um i guess we're here to talk about psychedelics and the 5ht2a and related systems and active inference and so i'll describe some recent work on this and looking forward to talking about it with you so let me get my screen sharing on okay coming through excellent so i'm using my uh uh active inference workshop slides as a base and i'll go over those with a little bit slower than i did uh in the 10 minutes i had there then we'll take it from there and talk it out this is you know part of an ongoing project to both try to understand the mechanisms of psychedelics but also the impacted neurotransmitter neuromodulator systems and the roles they would have in active inference and their potential significances for their use as machine learning parameters potentially when it comes to psychedelics um much of the attention has been on the 5ht2a system and that's a lot of evidence indicates that it mediates the effects um barely any work has been done in machine learning on this and there has been work done in active inference and i'm going to introduce some a compatible but slightly different take the potential significances of these systems within machine learning a good amount of work and with an active inference has been done on the potential roles of dopamine as a parameter encoding the confidence one might have with respect to policy selection diane and others have suggested that serotonin can act with an impotency to dopamine based on a variety of lines of evidence from neurophysiology and so some work has also been done in introducing a complementary uncertainty parameter in the form of serotonin this opponency can be thought of um almost yin yang like with uh dopamine being more of the yang and serotonin being more of the yin dopamine in general uh indicating in the neuroscience literature's incentive salience uh confidence and if excessive potentially impulsivity uh serotonin would be an opponency or an or complementarity i would indicate the satisfaction of desires states of uncertainty and potentially a factor helping to allow for patience rather than impulsivity with respect to policy selection very little work though has been done in machine learning uh on 5ht2a and also in terms of active inference in terms of actually uh modeling the

5hd2a system as an active inference parameter work is probably going to begin soon in this direction largely informed by the rebus paradigm which we'll talk about soon but go back to uh the fundamentals of these systems uh the work that's been done so far with the serotonin system can be thought of more as relating to the 1a system this would be in contrast to the higher threshold i believe it's a 5 to 1 ratio of sensitivity uh for the receptor of the two-way system so when serotonin uh both of these by so 5ht is serotonin and you have some i think 14 maybe 17 classes of receptors depending on how you count them but you know so how are we going to talk about all of them well maybe eventually it seems that we might be able to go a fair ways uh focus in terms of understanding their significance focusing on a few classes and sort of working our way forward with the additional receptors maybe having a common significance to the other ones but just because they're located at different parts of the brain it's a common organism significance but functionally you need them to do different things if located in different places into more detail soon but the 1a system is shown to in general have more inhibitory effects on the signaling of pyramidal neurons while the 5 hd2a system tends to be more excitatory carhart harris and nut have suggested that these both in a face of challenge would be responses to challenge but the 1a might be something more like an active or a passive coping response being more still patient stop and think as opposed to the 2a system which might be more of an active coping response in the face of challenge something more imaginative and potentially creative let's yeah let's keep going going back to dopamine versus serotonin though um serotonin in general not focusing necessarily on 1a versus 2a there seems to be some evidence suggesting that dopamine will push you or actually a good amount that dopamine tends to push the organism towards more externally focused attention in terms of uh helping to increase the gain on more exteroceptive and salience networks for externally focused attention 5ht in contrast seems to be up regulating the default mode and more internally focused attention so that's another interesting difference but to come to the 2a so these receptors as i mentioned before they tend to have high thresholds for activation by serotonin 1a will be modulating mood as their serotonin levels are going up and down you might get this sort of gentle adjustment but 2a it's going to take more to really get those systems going in fact some of their activity might be modulated by things other than serotonin so there's evidence that things like carbon dioxide um maybe blood acidity in general is changing their activity and uh maybe the endogenous dmt system which we could talk about later uh so psychedelics tend to come in about two classes uh one or classic psychedelics now one would be based on the structure structure of tryptamine the precursor of serotonin and the other would be based on the structure of phenylethylamine uh precursor of cachetolamines uh both of these seem to be active at the five hd2a system which a variety of lines of evidence have been shown to mediate the effects of psychedelics to a very substantial degree in the uh machine learning type work i've been collaborating with um jara shikhspahi to try to reverse engineer these different neuromodulators as machine learning parameters and so we've been using an architecture which has elements of hans nettuber's world models and uh danajar is a dreamer to interpret uh 5ht signaling as potentially being a pr as the different classes receptors as parameters for imaginative planning or an active inference terms a sophisticated entrance with respect to world models you'll see is this working yeah it's working so and in han schmidt hoover's architecture they'll use these auto encoders to take these different frames this is in a doom game they'll compress them into a lower dimensional latent space and then they will unpack these through this training of the autoencoder to learn how to reconstitute these frames you get this low dimension source of low dimensional vectors in latent space which can be used for training and with much better computational efficiency and some learning properties um when you don't have to go all the way back to pixel space where things can get messy and you don't know what's causing what if you work with this latent space you can come out substantially ahead in terms of training and so you have this variational auto encoder which is learning how to compress scenes so this is one aspect or compress the observations you then feed this latent space compressed vectors into this recurrent network which learns to predict the state transitions you know shove this into a much lower uh dimensional controller which uh they trained uh with uh evolutionary strategies and then used this for policy generating new observation and the standard sort of um reinforcement learning uh paradigm uh the interesting thing about these is they could also be run offline so they you're able to take these and uh take the outputs and feed it right back in and you can do imaginative rollouts so you can use this for planning or for having the agent train in its imagination and so this would be an example i believe the left would be but scene and or the left would be uh the the compressed latent space and then the right would be the agent actually i think the left is what was actually seen this is pixel space and the right is the the imaginative latent space it tends to be you're not really seeing it here but it tends to these imagine trainings like like imagination relative to perception or sensation it's less fidelity but you can still use it for the sake of planning and learning but one issue is that these agents can do a kind of um self-adversarial attack when they're training their imagination so if the the temperature the uncertainty isn't high enough they can uh basically come up with these sort of fantasies of uh easy environments where they you know are just winning but then this doesn't transfer to the real world and so then by introducing a temperature parameter to increase the uncertainty of your imaginative rollouts um this allows for better training back to the real world and so this um i've been suggesting is potentially one interpretation of the 5-ht-1a system um as increasing the temperature um putting the organism basically instead of dopamine would be you're confident go forth and act uh serotonin acting on the 1a system would say we're not so confident stop and think or imagine uh don't act right away but let's let's think this out or let's imaginatively plan this out look a little bit in our mind's eye before we leap the two way though seems like it's almost like a hybrid of the 1a and dopamine where it's like okay imagine more but it's not necessarily doing so under a state of uncertainty within these um imaginative rollouts things are appearing more vivid where the gap between imagination and reality uh seems to be uh narrowing and the idea is of if you're thinking back like passive versus active coping so you're uncertain let's say the 1a system is in the mix now you're okay i'm not going to overtly act i'm going to stop and think i'm going to reduce the gain on the um expected free energy or the expected value of pursuing a policy right now patience but if the 2a gets in the mix it's going to tend to be more active you're going to be imagining you're going to be more likely to act on these imaginings because you're uh they're more compelling uh more vivid now we're coming back to uh psychedelics but i want to lay out this foundation because the whole way through the method is basically a kind of marian neurophenomenology trying to understand the mind at computational functional algorithmic and implementational levels of analysis and then mapping this on to rich descriptions of experience of subjective experience so mapping standard the cognitive science multi-level stack to phenomenology so the machine learning is part of pulling off this mapping we could also or active inference is part of that uh to rehearse um the basics of rebus uh other am i coming across all right or just checking in okay so the relax beliefs under psychedelics this i'd say might be the dominant paradigm for understanding the significance of the five hd 2a system in creating the effects of psychedelics under a predictive processing paradigm according to rebus you'll get i don't think i have to rehearse predictive coding here but very briefly uh the idea is you have a hierarchy where of predictions where you are passing your priors or your prediction errors down the hierarchy back towards the primary modalities where the observations are coming in and then you're passing prediction errors upwards and this coding scheme is efficient especially if the events that you're tracking or trying to model are clocking slower to the internal states so with something like telecommunications if all you have to do is uh transmit what's different between the frames and uh i don't know 24 hertz used to be what tv was back in the day but if most of the pixels are largely the same frame to frame and all you do is you show what's different you get this energy savings and when you're dealing with an organ as hungry as a brain two to three percent of your body mass 20 of the budget that the savings could be substantial uh nature world of the quick and the dead natural selection might have picked such a thing within creative coding predictions are thought to be encoded by these uh so if you think of cortex as um jeff hawkins says it's roughly uh a dinner napkin and thickness an extent that's been crumpled up and shoved into your head cortex has these different layers um roughly six and this is distinct from like levels of a hierarchy but the cortex the sheet if you go like across the napkin there's these layers about six and depending on where you are they're more or less distinct but the l5 or the deep pyramidal neurons these are the ones that will leave cortex do things like form loops with thalamus the striatum uh cerebellum spine that's what will is thought to encourage your predictions these l5 deep pyramidal neurons not deep in terms of a deep police hierarchy but deep in terms of uh relative to the coracle surface more in there um and within the napkin uh the prediction errors are thought to be encoded by the superficial pyramidal neurons on layers two three so l5 is forming these big um synchronous complexes with the thalamus probably not the only way it does the predictions but this will be um aggregating information from multiple different places giving you and because you're able to get a a lot of different causes brought together it's going to be a good source of priors it's going to be a good source of your expectation and your modeling and then you use this to try to suppress the ascending stream of prediction errors along this superficial layer 2 3. you can go into layer later if that's useful but the idea is that with under psychedelics you take l5 pyramidal neurons these excitatory neurons and you excite them so much with 5ht 2a stimulation that they fall out of sync with each other and it's um it's a kind of paradoxical effect when they fire too readily they don't coordinate anymore in establishing a synchronous rhythm and because of this they're actually paradoxically relaxing the priors become less precise the uh belief or free energy landscape becomes flatter and so now the idea is with your expectations being less more information from the periphery is capable of penetrating deeper into the cortical hierarchy registering more in consciousness and impacting learning and updating to a potentially greater degree and so this is a model of the much of the phenomenology of psychedelics as well as their mechanism of therapeutic action for helping people to um as pollen would say like how to change your mind helping you to change and change your mind to update i think rebus is uh largely correct but i've been introducing a potential refinement of it um in something i call albus which includes in addition to rebus effects uh something i call sevis effects or strengthening beliefs and so uh the idea is that you might not get exactly this desynchronization effect along the whole dose response curve of 5ht2a agonism or stimulation of the 582 t2a receptor rather this might be the case of what you would expect from very high levels of agonism um that's really strong excitation but let's say we increase the gain like just a little bit or a moderate amount i suggest it's not at all obvious that this would cause the energy landscape to flatten this would not cause them necessarily to fall out of sync with each other in fact they might be able to sync up even more powerfully if you just increase it a little bit or moderately we don't actually know this might vary from you know organism to organism uh but the idea is we might need um different regimes different accounts depending on where we are with what level of stimulation before we would expect to see the kind of paradoxical desynchronization or potentially more straightforward enhanced synchronous ability yeah to bring it back to predictive coding so here if you're going to try to illustrate this uh the purple would be your deep pyramidal neurons uh supposedly encoding or that are thought to encode your beliefs and then up top you have the white circles this would be your superficial pyramidal neurons the upper part of the layer and this would be encoding your prediction areas that you're passing upwards and so you're seeing these i'm illustrating the synchronous complexes by these red swaths the predictions going downwards are these red arrows and then um i'm depicting basically the prediction areas as these gray swabs some and black arrows so within predictive coding there's been significance assigned to different eeg and meg frequency bounds as potentially corresponding more or less to predictions or prediction errors more specifically the slower rhythms like uh alpha and beta where alpha is maybe roughly 8 to 12 times a second betas maybe 13 to 30 times a second that these slower rhythms would be the ones encoding your predictions uh pass downwards but that your prediction errors might be more likely to be encoded by gamma frequencies you know anywhere from 40 to 120 even even higher uh oscillations per second uh so within this part of the reason that this might make some intuitive sense is if you're thinking of the formation of a synchronous complex of neural activity uh that uh faster would be smaller and slower would be bigger or more inclusive more integrative this might make some intuitive sense if you're thinking of okay so you're trying to get this agreement this negotiation of a common rhythm of a bunch of signaling units if you're having more units spread over a bigger distance you're trying to get them to go come into agreement this might necessarily have to be a slower process but if you're let's say like quantizing prediction error generally like a little packet of like a local agreement that you're passing upwards well that you can potentially clock a lot quicker um so that's the basic idea there so if we were to look at rebus it would look something like you see uh the the right bottom uh the swath of red is more restricted in scope and the arrows are a little bit smaller and so this is idea of you've relaxed your beliefs this is um if you get this paradoxical um desynchronization but uh and so once you have this you're seeing less of these prediction areas are being inhibited by this descending stream of predictions and they're able to make their way in more deeper into the hierarchy and this would be showing you know one modality obviously the brain is not just one singular hierarchy but it's a hierarchy multiple intersecting hierarchies corresponding to each of your modalities all brought together stitched together into a hierarchy up top would be illustrating a sebis effect and so the swath is bigger and the arrows are also uh darker and bigger and that's and more prediction errors are being inhibited there's an additional factor uh which is five ht2a receptors are also found on the inhibitory interneurons of layers two three and so in theory uh and there's some evidence i believe that suggests this uh this would correspond to a default reduction in the gain on the ascending stream of prediction errors or you would say an active inference speak um less precision weighting of the precision of the prediction errors and so they have less of a contribution of the bayesian updating that's happening at each level of this belief hierarchy and so that's illustrated by um slightly uh uh i don't know if this is yeah the on their own the arrows are a little bit smaller so the idea here with the cebus effect would be not only might you have stronger beliefs but these beliefs might be further shielded from contradiction by inconsistent sense data i think this potentially provides a fairly parsimonious account of some of the conditions under which you get certain perceptual alterations of a hallucinatory variety so for instance people will sometimes report hallucinations in sensory deprivation tanks they'll sometimes i think it's called a charles bonnet syndrome where as you start to let's say go blind or deaf you'll start to get like auditory or hallucinations within the modality as you lose input from the external world uh this also you could think of at a higher level i don't know if this works but you could think of potentially delusions in a similar way a belief which is overly strong and resistant to updating from uh potentially uh contradictory and potentially correct sense data so in theory for talking about sebis effects you might have a single parsimonious account for both hallucinations and delusions that's incorporating a little bit more of the neuroanatomy that being said i'm not saying at all that rebus is incorrect i actually think it's probably very correct and uh maybe crucially important for uh explaining some of the really powerful changes that people observe with psychedelics as a therapeutic intervention get into a little more details here and it's okay uh y'all with me still there okay cool i mentioned a header here earlier so i've taken these um predictive coding hierarchies here and i'm just illustrating them as these triangles and so here would be let's say uh you're uh i think in this framework of albus i'm suggesting uh we need to even uh in addition to including things like sebis effects i think we're going to need to take on things like basically consciousness higher order consciousness to actually really adequately explain what we mean by relaxed or strengthened beliefs and what actually is happening with these uh different neuromodulators and maybe attempts at creating analogs uh here you know up top would be somatic sensation to the right would be your vision and then your hearing and we're going to try to bring all these things together and so if you're thinking of some sort of hierarchy like here you might have like small beta complexes you can think of each of these as like a small modeling effort and a small belief over whatever is in the scope of this synchronous complex where the idea being that when you synchronize neurons this allows them to be aligned in their activity and time such that their signals can sum and be integrated rather than decay and so uh this would allow you to have potentially a joint belief over whatever is in the scope of a synchronous complex uh pascal freeze calls this communication through coherence and uh for a reference for this idea of like an inverse hierarchy of size and speed butsaki would be a good reference for that i'd say so here we are on the left the beginnings of modeling of increased hierarchical depth with small fast beta complexes the extraction of things like line segments and maybe basic contours and vision uh basic uh patterns within sound within touch but then let's say we nest these within maybe maybe slower broader more encompassing beta we could have more hierarchical structure for our modeling effort and so here you might start to get the beginnings of things like objects as you bring together these different features the beginnings of things like objects with respect to touch your your or rather your own body in relation to those um the beginnings of things like meaningful sounds and so then let's say we bring all the modalities together now we let them cross talking about multi-modal representation that you've organized according to a coherent perspectival reference frame well now you might get the beginning of something like phenomenality or experience and so here there's a blue swath corresponding to alpha frequencies here um at alpha frequencies that would roughly be it looks like the size that you would need so 8 to 12 times a second of a frequency of updating or or sampling or estimation that this would be big enough to get you the entire sensorium all brought together bringing together that the different hierarchies into a hierarchy alpha has some additional significance in that it has particularly concentrated sources of alpha power from midline structures i take this as being important because if you look at something like the posterior midline we look at let's say the posterior cingulate cortex which is heavily implicated in the psychedelic and also the meditation research literatures as when alpha goes down there seems to be some associated with phenomena such as ego dissolution the idea that alpha might be important for coherent egocentric perspective in addition to bringing together your sensorium for this mutual information these midline structures in particular the posterior cingulate um there's this old idea of the papez circuit where it's this uh core circuit for emotional memory um that would roughly go manulary bodies around the brainstem it goes up through the mammolothalamic tract i think anterior lateral thalamic nucleus goes through the cingulum bundle posterior cingulate into the hippocampus out through the foreign inks back to bodies don't worry about those details the important thing about this is uh this was interpreted um by um actually i said papas i think it's papes but it was interpreted by papes when he was etching out these elect sources of epilepsy form discharge as this core circuit for emotional memory and part of the reason that and so the importance of this is uh the mammary bodies are receiving stretch receptor information from the neck and so this is going to be potentially important for knowing where your head's pointing especially if you're talking about like which will be important for us and maybe even more so if we're talking about like our inner rodent or or whatever that you know where your head's pointing is going to really help for bringing coherence to explaining your sensorium uh also at this islamic relay you're also getting inputs from the vestibular apparatus of the inner ear and so this is giving you like the yaw pitch and roll and so not only do you have your entire sensor in all the information at these posterior midline structures before they funnel into the hippocampus as a source of new memory not only do you have the entire sensorium brought together you're on top of it all in network terms but you have the information you would need to organize it by your egocentric perspective this yeah so here we go predictive coding we covered uh sebisfx redus effects and now i'm introducing like a basic model of uh perceptual synthesis where we have something of a cartesian theater here enabled by um uh the ability of slow rhythms of a large enough extent to bring together the right kinds of information you would need basically create a coherent uh world model so y'all with me stellar uh in this preprint i describe how we can maybe think of with respect to either sensation or imagination a different combinations of rebus and sebas effects happening potentially corresponding to different levels of stimulation of these receptor classes of the 582ar in particular by the way um i think 5h 5 ht uh 2c that's gonna that's one to watch well it might even be like almost more strictly rebus in its nature but that's something to talk about later um so for sensation on the left here you know this would be there's a person out in nature and they're having an interaction with a tree and then on the right here would be a similar thing but this is imagination uh maybe you know with your eyes closed maybe not but this is you know um here you're imagining yourself uh and interacting with the tree now you know imagination it doesn't need to be in a third person but i'm just taking advantage of the fact that we're talking like if you are to think of yourself in a third person if you're going to move from this um was it the as james called it like from the eye to the me uh to this objectified selfhood that part is going to require imagination you can imagine yourself from the first person but the third person point of view um you're gonna require um imagination i guess that or a mirror or a good setup of mirrors maybe you could do it that way so that actually might not that might be non-trivial but on the left here so the top we have let's say unaltered if such a thing ever exists we're always altered in some ways we're always on some our brain stems pharmacopoeia is always dripping some combination of drugs to us it's just uh sometimes it's a very different regime and you know we say it's altered but there is no like singular baseline you know we who knows how much we both vary as individuals and across individuals along these lines so as we move down let's say we were talking about a revis effect and let's say you were getting a rebus effect with sensation so in theory if your perception is entailed by the predictions then it could be the case that things might seem less vivid to you it may not be the case that your perception is more vivid i don't know if i would say this is strictly um entailed by the rebus model but that's one thing you might think potentially if these slow rhythms if some of these are actually entailing your conscious experience if these are reduced in their coherence and their power in theory you might get a diminution of perceptual vividness however in a cebus interpretation things might uh appear more vivid not necessarily more vertical but more vivid it seems like a rebus interpretation would say that things might be more detailed more accurate um more richly textured with more faithful correspondences to the external world when you're perceiving but potentially less vivid not because this is a strict entailment but that seems like one suggestion under a cebus interpretation more vivid but not necessarily more precise it actually might be a little bit smoothed over like like an auto tuning or a filtering but um also though uh more compelling potentially and so this in theory though would be the phenomenology of micro dosing in theory when i get into and so i try to illustrate this with the brain you know bigger or smaller synchronous complexes here um so albus the idea is well we could have an admixture it's not just one set of beliefs but we have a hierarchy of beliefs and in fact a heterarchy so maybe the alpha starts to get pulled back and that loses coherence and that seems to be one of the strongest associations in the psychedelic meditation literatures is alphas what people will emphasize is some of these uh ego decentering or dissolving effects but maybe beta stays the same or even increases in power and so under a rebus model for instance uh the idea would be that you know things like alpha go down in power but things like those particular smaller beliefs they can come together in different ways and different sort of creative combinations and so you might get something like synesthesia with psychedelics as like a creative recombination of things um under this more anarchic state once you took the the alpha self ruler and you reduced its its tyrannical grip over your perception and so this more free-flowing perception you get all sorts of different creative combinations uh and consciousness and so yeah so that's one thing is trying to basically map different combinations of rebus or sebis effects that you might expect under different doses with respect to either sensation or imagination so really digging into the phenomenology and using psychedelics as an opportunity actually in a way to indirectly test models of consciousness but also if we're going to explain what these substances powerful substances are doing we really actually need to take some of these questions head on maybe even the hard problem some of the more so in this pre-print um i also are getting into okay well what do we mean by beliefs what kind of beliefs are we talking about and so uh so one kind of beliefs would be so you know here you might have the beliefs with active inference and the free energy principles beliefs the whole way through oh which belief swimming and so um you know and rebus is saying what's specifically being is being relaxed is your deeper beliefs your internal working models the higher level beliefs corresponding to things like selfhood and its relation to the world um things like autobiographical and narrative selfhood these are the things that are losing their grip and within rebus you know i don't wanna like there is room for sebas effects and and some of the ways it's discussed where it's not just across the board all the slow rhythms are going down the idea might be that you know i've heard it discussed differently and you might get you know an indirect strengthening as you remove these slower integrative and potentially suppressive processes but in terms of types of beliefs if we're going to be talking about cognition or if we're talking about something as like highfalutin as the self now we might want to say like well what kind of self are we talking about like we're talking about like a minimal embodied self are we talking about an extended self and even those like there's all sorts of bad questions like what goes into the minimal embodied self how it's relating to the extended self and how many maybe there's different kinds of those and so if we're talking about the construction of self in its relation to the world um and objectified selfhood the ego the the self as construed over time and across circumstances we're going to need a fairly rich account of cognition and so here i'm providing an account of sophisticated affect of inference or imaginative planning and policy selection as orchestrated by the hippocampal and entorhinal system and so this is uh drawing on some of the work um doing with tim verbellen and others i'm trying to decipher that system but basically uh you know you seem to have uh so here you know hippocampus can be thought of in a way as at the top of the cortical heterokey when the prediction errors are not so you as the frictioners are percolating upwards being contradicted or stopped by predictions you stop them as low as you can as close to the modalities as you can that's the maximum savings and that's what you would expect with the maximally efficient prediction and learning um or if you can't predict a pattern if nothing the genuinely novel that makes its way all the way up the hierarchy into the hippocampus as this prediction error storage place of last resort and you this then can be stitched together to create memory and specifically what hippocampal and enter rhino system seems to do is it encodes this sequence memory it has multiple functions but one of the ways in which one of its core functions seems to be the encoding of trajectories of the overall organismic system through time and space and so if you look at the work of people like reddish you'll actually see for instance as this rodent is like in a maze um figuring out where it's going to go you'll see these sweeps of activity within the hippocampus kind of play out in different directions across these place fields or this mapping of space and then the organism will take like one branch or another of these uh potential sweeps of predictive activity and so uh i probably don't want to get this here but i have an account of like the anterior versus the posterior hippocampus and be qualitatively different where the anterior hippocampus seems to be the one that's more predictive giving you these sweeps giving you things like counterfactual processing the anterior seems to be more closely coupling with the the posterior censorium and will be more important for more specific episodic encodings and maybe more faithful episodic rememberings the idea is you have these little hexagons which corresponding to the mapping of space these hippocampal place fields where is the organism within some sort of space and then they're playing out in some trajectory and so now along this trajectory it is you're having the prediction errors from the cortical generator model which can by pointing back to cortex can unpacked unpack at every point in space um the experience you would expect for that particular temporospatial trajectory and this would be an account of episodic memory and episodic imagination or the stream of consciousness if you will okay so here we are and i think yeah this is the last thing i'll so here we are with these trajectories being played out and different experiences being entailed at each point as it plays out and they have a given length it seems that there's um a extent of this um i think james calls the specious present or the idea is there's like a thickness of the moment we find ourselves in or this might be a somewhat different idea but there's there seems to be three seconds is seems to be roughly the upper bound of what we would call a given extended moment of experience um edelman i think called like the remembered present where there's a little bit of like a prediction and a post-diction and you're moving back and forth in this sliding window frame of experience and so one interpretation and so within here as you're doing these different episodic rememberings and imaginings that one thing that the 580 2a system might do both increase the vividness of what you're imagining or perceiving as you go but also allowing you to have more extended uh rollouts before you have to reset where it seems like a roughly three seconds might be the upper bound basically i'm i have uh within the preprint this uh this comic description this neurophenomological comic where as you're increasing the dose imagination is getting closer to perception and you're getting a longer runway to work with for this extended present moment with within which you are planning and learning and making sense of things and so it gets a little bit bigger with the micro dose a little bit bigger with this threshold dose you know things are even more vivid you know when you started out you're just like kind of moving through the world i'm kind of awake and kind of conscious i guess and then it's like i am like things are crisper and a little bit more creative like not that much here but a little crisper um this would be something like i don't know like uh someone does like a programmer in silicon valley like micro dosing is very popular it doesn't seem like they're trying to like trip but they're they're almost using it like like like a little bit of a a stimulant trying to like increase absorption whatever they're doing so as we move through you increase the dose some more and now it's even more vivid and more creative more different imaginings you're having access to more more is manifesting in mind but then after a certain dose it seemed what i'm illustrating is that basically the the coherence starts to go and you start to get smaller rollouts they're even more of it than they're more creative but they're becoming uh less uh clear in terms of the sequencing they make less sense with respect to achieving particular goals and so i think something that i should explain and we might have to talk it out but the the underlying this is it's okay so uh these hippocampal resetting events are occasioned by different things one is if i go from one room to a next um if some sort of like really surprising thing happens that's gonna trigger this um sharp wave ripple activity and a re-tiling of space within which you're doing your sophisticated inference um this can so something either uh so there's there's some evidence that reward prediction errors if something really really good happens that might cause you to basically ramp up the hippocampal activity encode that you know generate like the phasic bur help generate a phasic burst of dopamine learn it and then uh keep going or if something really bad happens this might be an opportunity to pivot what you're doing is not working let's retile let's try again this will be a function of both things like what's the neuram modulatory tone what are those like the circuit level conditions like maybe directly what's your level of 52a simulation that might influence uh how long you get before you have to reset but it could also be explained at the level of experience itself at the level of the expected free energy changing as this emerging cause so as i'm getting creative if i get so creative that i'm no longer making sense of things my expected free energy if that goes up like that spikes up because what i just said is really surprising it's it's it's novel it's experience vividly and it's not making a lot of sense that might be something that would occasion this resetting and so the idea is a model of basically the stream of consciousness or episodic memory and episodic remembering and imagining which becomes uh coherent more rich more absorbed within this first regime and then a second regime that's more properly psychedelic where very un it becomes extremely creative so much so that it's um becomes qualitatively different than normal experience and even so much so that like for instance at a certain point you're not going to be using this to enhance your job as a programmer you will be doing this on vacation with trusted others or something like this hopefully and this is not something to take to work that you know people are not mega dosing at work hopefully or they will not be working there for very long at most workplaces yeah that's basically and so then coming looping back around to the beginning the ideas with a deeper understanding of these systems that includes some rebus aspects and some sebis type aspects we might be able to use these as inspiration for novel active inference and machine learning parameters [Music] potentially you can think of like getting stuck at a little like people you know if they're living like a life of quiet desperation or something they've gotten stuck at some kind of local maxima and they're looking to change things they're looking to get to kick out of it they're they're not finding an elegant path to minimize their expected free energy and so if you've found yourself trapped at this local free energy minima you can use psychedelics to turbo charge your sophisticated affect of inference enough to maybe find a novel solution see something that you would otherwise and then i guess the final thing i would like to point to is in terms of using psychedelics as a probe on mind you might have different interpretations of what they reveal if you're thinking of either a strictly sebas story a strictly rebus story or some kind of combination of the both of them and so i'd suggest that for instance you can think of a good amount of psychedelic phenomenology as corresponding to uh the unmasking of your core priors and so things like let's say fractal um fractal phenomenology that maybe would be a core perceptual prior maybe even partially evolutionary partially as a reliably learnable posterior or empirical prior for making sense of the world and this would make some sense in that if you use fractals as you're as generating your predictions that's a very efficient basis set because you can you can do a lot of different combinations of fractals and this is especially going to be efficient if that's what nature itself is doing you're using different combinations of fractal processes to model a world which is largely structured by different combinations of fractal type generative processes further things like entity perceptions uh associated with uh different psychedelic experiences um archetypal type experiences those also could be thought of as the unmasking of core priors potentially strengthened core priors let's you know because you'd expect from active inference that you know entities is something the kind of thing you would as a a very powerful empirical prior because that's how we survive especially agents such as us who think through other minds and where our primary niche is largely each other whether we're you know so the details of psychedelics might actually reveal these foundational inductive biases or foundational lessons in the learning curriculum and the structure learning whereby we come to be the kinds of beings we are and that is it for now thank you adam that was super awesome um i am like particularly overloaded right now on like my 1a receptors which is like i just want to imagine and step back um before i ask some questions i know daniel asked a lot so i'm gonna let him like ground me a little bit in reality and have the first jab i was just writing a bunch of stuff in the live chat so definitely anyone can ask because you jumped through many areas so if anybody has a question they can type it but let's take a quick breath collect our thoughts and then blue you can ask the first question wait i just passed to you to like ground me all right so so um i guess i'm already here um all right since i'm already here i'm gonna like unclearly ask a whole lot of questions at once and also like try not to fangirl out too much because this line of work is really super awesome and unifies a whole lot of my particular interests um so just really quickly what um do you think the relationship is if any between inference and memory and can we quantify that through the brain the brain system dopamine serotonin like do you have thoughts on that i'm sure you do or do you have thoughts that you can summarize in you know 10 minutes or less so in terms of memory um there's a sense in which it's all like there's a sense in which within the free energy principle and you know all like was it all effective causation is inference and it's all inference based on different kinds of memories so your belief every belief is a memory and everything that happens is a kind of action as a kind of inference but there's another sense though of like memory of the time i'm the kind i'm trying to get to like the focus on the hippocampal enter rhino system this orchestration of the overall memory system and this encoding of the general genuinely novel in these episodes and these frames of sense-making this a more complex kind of inference and a sophisticated one specifically where you are inferring the most plausible a conce rather than so so there could be a sense for ins for instance in which you might get a fairly under some circumstances maybe for like recent things that that that aren't like uh i don't know what the shelf life would be but like so some of the literature and a couple weeks couple months i don't know but like that actually the hippocampal system through just this chaining of attracting states along these trajectories could point back to cortex and play out something that's actually like a fair a decently faithful but still very biased account or a reconstruction a remembering of what occurred um but for things that are older that's probably not the case even for things that are newer there's this constructive so it's always constructive and that it's always a kind of every moment perception is inference at every moment there's a creative aspect of filling in but actually like situating yourself within some sort of causal unfolding that's going to involve a lot of value cannalized policy selection that's going to involve you using constraints like well what kind of being do i think i am and what's plausible and that's going to determine when you're going down this like bran this branching forking pass of different possibilities like you'll be moving between like well what might i have done under this or that and you'll be picking plausible accounts of you partially as you go and so some remembering might be a little bit like the naive like playback one if you play back the trajectory it's got pointers back to cortex and then you have this belief hierarchy and so you give the prediction errors back to where things were new and then it just unpacks itself as a good enough account of what happened at each point as he went through some trajectory and it's a decent semi-faithful playoff but there's also going to be this other kind of memory though that's that's not faithful at all and can really deviate a substantial amount of well we're we're in hard to distinguish from imagination and continuous with it it's like how do you know if you're remembering something or imagine you use these different cues one cue you might use is perceptual vividness so you would assume like if i remember something it's gonna be a little bit crisper than if i just like totally made it up uh the other one might be like just the plausibility like yeah that's like the kind of thing someone like me would do and but you can get it confused and that would be a source monitoring error and i think that's actually in a way for psychedelics um both in terms of this core process of figuring out who you are where you've been where you're likely to go and where you ought to go thinking of these pathways as what it is to be an agent as revealing ways that agency and different self-consciousness and self-hood can vary but also in terms of this i think needs to be brought in in terms of what we might expect to be impacted and what we might want to prepare for and heart and harness with things like psychedelics so for instance there might be a very real risk for phenomena such as false memory so if the gap between perception perception and imagi and imagination starts to narrow if your imagination starts to seem more vivid it might become more difficult to distinguish between a remembering and an imagining or um yeah so things of that nature so i think the issue of what is memory um in its relation to inference is going to be key for what it is like we want to alter and what it is we maybe don't want to alter but might accidentally alter um when we do different types of interventions so like you played so perfectly into my follow-up question was like how does creativity play into that right and like what is um maybe the function of creativity like i mean personally my autobiographical memory is awful but like i have a really stronghold like good factual memory for like all kinds of ridiculous things so where does creativity play into that memory perception inference like paradigm and are like the people that are more creative or that take more psychedelics like do they have a less like accurate memory or are they more likely to remember remember things inaccurately or even like what is inaccuracy sorry like let me just get super meta because i really fundamentally believe that perception is different for all of us but like what is written in my biochem textbook less so right so maybe that's why like i put less priority on my autobiographical memory because it's i believe it's functionally inaccurate but like whatever how many angstroms are between each subunit of dna like that's way more you know important right i mean that's one of the hardest questions i can think and sorry one of the most important too um so there's this i'm trying to forget the name of the paper it's this dual constraints model of creativity um that does address like head on the issue of kinds of creativity and psychedelics um i'll post that uh later it does seem like for instance there are i wouldn't say necessary trade-offs with respect to uh different processes as to whether they allow for realizing different types of value so let's say so with memory or with creativity you might distinguish between divergent and convergent creativity so like conversion creativity where like you want to hit a particular target and generate a novel solution for a particular challenge or divergent where let's say you just want to think of something new um maybe just a novelty generator from which you then might select later convergent solutions in a longer process or just uh as part of an open-ended exploration you know there's different ways like you for instance like you could get if you just turn up uh the heat the temperature and you make things more imprecise and unconstrained you might be able to get all sorts of different combinations and this might be a good like fountain of diversion creativity some of the evidence suggests i believe that uh even with the micro dose regime so i think people who are doing like micro dosing usually they're looking for things like enhanced conversion creativity i don't think the evidence is great that that's actually being achieved but uh divergent creativity might be more likely to be associated with psychedelic experience proper of combinations of things that just you never would have imagined it's not clear it's not necessarily even useful for any particular thing it's not hitting any particular target necessarily it might be after the fact it might reveal something to you that you never imagined that made all the difference maybe like it juked your defense mechanisms in terms of like your policy selection was leading into garden paths you're seeing some things and not others and then it took the blinders off it revealed something to you that you might have missed but i'd say that people definitely vary partially maybe due to things like low-level parameter tunings like different people will vary in terms of like the natural level of the gain on these systems the natural tendencies for occupying these different regimes and so you'd expect that to vary either through some combination of things best explained at low levels like you know different alleles of neuromodulators um and some of it would be not well explained at that level it's more just that's the way the structure learning happen that's the way the system bootstrap itself different kind of agent and and everything in between one thing that i find to be like uh i want to go into is that humanness in general us as a creative species that these pathways might have been important for allowing us to have be more creative and being able to combine things in a broader range of ways and that this potentially uh you know unlocked a more flexible type of cognition maybe a greater capacity for analogical reasoning the kinds of things that might have potentially precipitated this great leap forward and the birth of cognitive modernity where we became a symbolic species and cultural evolution came off the ground there there's an uh older idea by mckenna called the stoned ape theory that like there was actually like a period where we were in this symbiosis with mushrooms and we're eating psilocybin containing mushrooms and this helped to get that going i don't think we could rule that out but you could also get something very similar with a you know a couple high impactful mutations along these pathways that doesn't involve a mushroom eating phase and so there still might be a really interesting account of actually the secret of our success as a species as a particularly creative kind of hominid who's able to bring things together in more um novel imaginative ways that um could be involved in being continuous with the story of psychedelics and how they work and in the present day so that obviously you know i'm extremely interested in and also as a source you're saying individual difference like you know you know are there more like to this day more shamanic personalities are there you know are there more and to what degree can this account be told at the level of genes or is it or epigenetics or is it not well told there uh one yeah so that's those are some thoughts on that okay so wait now you just like walked into this trap that i laid for you so i've now got to ask this next question so you talked a lot about um psychedelics and sensory deprivation and now you want to talk about like our symbiotic relationship with fungi and i'm a huge like terence mckenna fan so if you've read the archaic revival which i'm sure you have so where do sensory deprivation in conjunction with psychedelic use like where does that fall on this like perceptual perception action mental action alpha beta brain wave serotonin dopamine spectrum like i just want to know your thoughts sorry not sorry but i'm sorry i don't know if this exactly gets at it but it's possible that um this would be rather different but like okay like this is this we're really going to go into conjecture land here um but so i think what the book is uh it's a recent one um about us is so that there's like two domestication books recently about humanness so one of them is uh the goodness paradox and us is a self-domesticated species then there's this other more recent one but the idea is that this um pathways if it depends which ones we're talking about will be quite different but they are up regulated unless they have a 1a variety you could think of that as being more conducive to something like a domesticated phenotype if your dopamine is higher and you're you're having uh you're expecting you're more confident in your policy selection then that might also cause you to be somewhat impulsive and that might make you not necessarily be able to discount utility enough hold back think things through a bit more and you um if the dopamine is too high less uh cooperativity if it's and so you can think of the serotonin to the patients that it's the 1a in particular as being um of a domesticated variety make and things might appear for instance less vivid perceptually um and imaginatively in theory along if we're just talking about those the 2a you know for all this um there's so much we don't know so so much of this is conjecture but the 2a does seem to have a little bit more of a dopaminergic character to it and actually it seems like that pathway evolved roughly around the time of the cambrian explosion um around with the advent of jaws in the context of predator prey arms races and so that might be consistent with like a dopaminergic significance so this like temporarily like increasing um your confidence your policy selection like turbo charging your action selection a bit so you can go out and get the goal of either getting your lunch or avoiding becoming lunch let's say you're moving into greater gain on imagination let's say we're thinking serotonin either of the one air 2a is doing this this then might compete with more externally focused perception sensation and so you're imagining more but because your mind's eye is occupied more with your imaginings you might be attending to the details of the environment a little bit less and so uh a species or an individual who might have to gain for whatever reason higher on these pathways um you might call this a little bit more shamanic or maybe a little bit further on like a schizophrenia type spectrum um might be perceiving the world slightly less faithfully their imagination is now freed up to be more freewheeling and more creative and this is allowing for different things to come into being so very speculatively we might think of like neanderthals as like less domesticated humans maybe even a little bit more like autism spectrum but like better able of perceiving the world precisely um and us we might have been a little bit more shamanic a little bit more uh although we are the dander cells also because we all interbred and the very you know the variance within is probably bigger it's definitely bigger than the variance between us and my the our ancestors and us um we we might be somewhat somewhat less i don't know i don't know if i'd say less dopamine i don't know if i'd say that so that's what i got for now thanks adam um great presentation just wanted to make two quick notes and then ask some questions from the chat so the first is there's no free lunch with strategy so there's niches that are just unwinnable where the phenotype cannot win and then there's other niches where potentially a broad range of phenotypes all can proliferate so whether or not a given strategy is successful or under what cases a given kind of neuromodulatory state is good or bad by what metric there's just no single free lunch or preference for lunch food then the second point was the complex gene family point was very interesting and you mentioned some other receptors in the serotonin receptor family there are many receptors including ones that don't have nice clean names like number one and number two and number three so at least 14. yeah so how it's exactly investigating function in these massively interacting and often redundant like if seven does something but only when eight is gone what's the actual mechanism there so here's some questions from the chat so first question from a chat stephen asked interested in the approximately three second window of conscious awareness adam mentioned does that trace back to william james stream of consciousness awareness and the temporal scale scales of experience i believe so so it's kind of like a three second echo is our experience or what does the three seconds represent so um so the empirically established phenomenon is there seems to be stitching things together into one continuous unfolding there seems to be a three second upper bound within this continuous unfolding i believe there is relevance to james's species present and edelman's remembered president of this idea of a kind of anticipatory prediction and looking ahead and a postdictive going back to you're moving back and forth of yourself in this unfolding and the speculative part is um i am suggesting that this might correspond to hippocampal endorinal tiling of space either a physical space or a conceptual space within which you're doing this and that the reason this upper bound would be the length of time you have before you need to refresh the attractors and re-tile speculative and then and the idea would be that this would correspond to something like a these sharp wave ripples that you'll observe we don't know for sure it's a perfect lead into this second question and then i have another comment from cambridge breaths so what you just said about ripples so they wrote wonder if investigating cortical traveling waves via neural mass biophysical modeling approaches in particular the work of wilson cowan can contribute to this work if yes how many thanks for elaborations timmerman um in collaboration with robin uh card harris has done some really interesting work along those lines i believe it was with dmt looking at traveling waves um i believe i believe in the alpha frequency as evidence for the rebus models and so the idea would be that if you are actually relaxing your belief landscape a reduction of like traveling waves of predictions going down towards the periphery and maybe greater inward going prediction error waves and uh some work has been done along those lines already by timmerman and others that's really and as i was saying before so although i'm like introducing additional factors i do not think rebus is wrong i just want to add a couple additional details some of the contributions from modulations of cortical microcircuitry might be different you might be able to get rebus effects from a variety of ways not just from this desynchronization but before i forget earlier you're saying about free lunches absolutely and i think this is part of like you know for instance like let's say you know if i would just crank it all to the max um if each one is like has its own function well you know it'll be depending on the nature in both generally and locally so we're talking about dopamine too much dopamine you might be impulsive too little you might not strike iron's hot and you might miss opportunities too much 1a or too little 1a you might be irritable and impulsive too much this is opposite problems 2a too much um it could generate over time a kind of psychotic predisposition that would be a loss of a free launch the one more thing that you're mentioning is okay yeah it's not just these two receptor classes uh there's actually you know 14 conservatively and one thing i'm wondering though is whether you get kind of like this is potentially wishful thinking but potentially not um you get the lie and share by considering a couple and that there might be like an order in they evolved and a ubiquity where the ubiquity um would be one sign for like how important it is to figure these things out so you know the just the broader the distribution in theory the more you get from explaining the ones with the broadest distribution and so the 1a and 2a have this extremely extensive cortical and subcortical distribution um but that's not necessarily the case like you could have for instance at pivotal areas of the brainstem you know you're right on top of the core value signals you know one class of receptor there that could be the difference that makes all the difference another aspect the hope that you can you know make progress incrementally you know with the goal ultimately you know we're going to want to get every single last one of these receptor classes is that the other the additional ones might have a common organismic cybernetic significance they're different because of their different locations and so that their functional account might still correspond to let's say overall levels of serotonin overall levels some neural hormone or blood acidity or some other homeostatic state or state of and so that the active inference significance might be the same but you're tweaking differently in different neural systems so that the overall coordinate effect is coherent with respect to um you're expected for energy with respect to realizing organismic value or minimizing prediction error with respect to the prediction of you doing the things you need to do as a successful being and so one more thing though is i'll just throw off the 2c idea is with these gene i don't know how far we can go with this but with this idea of like so with a gene duplication event once you have two copies of the gene this allows you to basically take one of the copies because you've got a spare that's still doing the same thing and now this other one can mutate and can separately be optimized by evolution to take on a different function and so the idea is that there might have been a common ancestor that looked a little bit more 1a-like but that at the extreme levels of it during this gene duplication event this became the 2a receptor and maybe this goes one step further to the 2c i'm not sure but maybe we go a couple steps further with this kind of gene duplication story and what we're corresponding to is peeling out and separately module in a modular fashion optimizing for particular regimes of organismic are there gonna be 14 to 17 i'm skeptical i don't think is that i suspect it's like maybe three or four top stories of this kind and that the other receptors it's these core two to four stories probably three um maybe achieved by having different functions locally but in their coordinated activity the same story globally nice there could be a lower dimensional manifold that gives a lot of resilience and also evolvability for the system and learning and one thing we've talked about before but it's interesting to think about is in your presentation you did a really awesome job of connecting the general kind of dynamical systems and inference questions to specific human regions and then it's interesting to ask okay well insects have a different brain layout they don't have the entorhinal so we can juxtapose cognitive systems and learn a lot about which attributes of the computation are necessary and sufficient so i wanted to ask which cognitive systems you thought were interesting to study in that sense i i have an intuition and a hope that may or may not pan out that the extent of convergence um is fairly substantial now things can i think flip in all sorts of ways you know i wouldn't just assume this is correct but let's say you know we're talking about within an insect um you know we would potentially look to the central complex and you know which which has a lot of hippocampal and to rhino functions and so and it might actually be a an even better system to study in terms of you would have you know complete access theoretically to the whole you know you'd have a much richer access to via something like calcium imaging um it that could be looking at something like the attractor dynamics of like you know so there's there's experiments like let's say like take fruit flies they put them on top of like a little styrofoam bowl and hook them up to virtual reality and they have them like doing different things you look at the central complex while it's doing that under different states of neuromodulation and we might see common organism significances and a common computational account in terms of the map mapping space and policy selection now how far back does this go some of this logic might even play out at the level of individual cells and like either like for instance like 1a versus 2a 1a so low levels of serotonin seem to have an effect of strengthening cytoskeleton dynamics and so that would promote like an organism kind of staying in place high levels actually destabilize the cytoskeleton and so they'll actually like literally create plasticity and fluidity of the morphology and allow the the organism to move around more and so some of these accounts might go all the way back to the like near the origins of life and individual cells and metabolism uh how far we can go i don't know and so some of these might have even been repurposed for the different social ill insects communicating although you could see these things flipping because you know what goes in the individual doesn't have to be the same in fact it can because you have things and one significance on the individual level that might mean that has the opposite significance at the group interactive level so um but i think it'd be extremely rich to like i my kingdom for that science to be done blue do you want to ask a question i will jump in here actually um you know when we talk about the origin of life you mentioned earlier in your talk that third person perspective is related to imagination and creativity and this goes back to kind of what i was asking earlier is it so much that the third person perspective is related to this kind of like coming outside of ego and looking at yourself you know from this kind of meta perspective or is it just fundamentally the dissolution of the self like how much does an ant think about themselves or like a bacterial that's part of a micro like or a biofilm like how much of of that is imagination like looking at themselves as part of a whole or that's just like myself doesn't matter i'm part of this bigger unit um so like if you're part of a well-coordinated collective there's a sense with that within active inference you will have converged a shared generative model for your coherent inaction that you're taking part in and so your role of what's being enacted there's a sense in which um it's not exactly like on some level in some way there is information of some kind which is not necessarily egocentric in fact it's like your role aloe centrically within the thing that like even if on some level it's like what what you are you as generative model if you were to give the generative model generative modeling process as a whole of perspective it's not egoic it's not from necessarily from a first person point of view even though the it it's it's from the the coordination manifold of the of of the group that's doing this this common thing am i kind of getting it yeah yeah that's and it's interesting like developmentally if we were to kind of take when these different perspectives of different kinds come online like for instance right out the gate it's not clear to me that we have like for instance coherent egos like it's not clear we attain to this like right out the gate when we're born and so there's like a sense like by default the you know the infant might be egoless right it's it's and by default it's doing this more like inactive coupling with its caregivers and this sort of thing and as not yet actually being um enslaved to an egocentric point of view there's kind of an interesting thing with language where it's actually like the ego itself seems like when we use it like in like psychodynamic terms is actually so we say ego central perspective it's like the i and it's the first person but when you say like ego as like a kind of like extended unfolding of you that's in the world and all like your possessions and your sense of yourself within a social context that actually is more third persony so it's like ego and from a such perspective is more first person even ego as like psychodynamic concept and the kind of thing you might want to modify with a psychedelic is actually more of a third seems like a little bit more like a third-person point kind of thing but um and so now the question would be though like maybe there's like an implicit um aloe centricity or non-egoic-ness being enacted by any coupling and maybe that's like the primal like the primal the primordial state um and then you attain to an ego central perspective but there seems like there's like another kind of um more psychodynamic ego or objectification of self the third person um of taking an explicit perspective they think it's it's actually first person and that you're perceiving it from a point of view but it's fictitious and you're taking the view on yourself as if from without and that might come later in development at some point maybe through like enactments of um mirroring between uh infant and caregivers or between con specifics but like if you're mirroring with others then you might be able to establish these cross mappings between what you do and what someone else's does and this could potentially help you to create representations that you can then harness for viewing you as if from the outside when we're talking about so when we say self when we say ego like what kind of ego what kind of self it seems like there's a primordial like minimal self egocentricity of just seeing from from behind your eyes but then there's more like a psychodynamic ego uh slash me uh which is involving the objectification of you adopting a perspective on you imaginatively as if you were viewing yourself from without and then contextualizing yourself across space and time and in different contexts different social unfoldings as what you would need for something like an autobiographical self something of that nature well so how does an ant see it though does an ant have like is the first person gone or like does is an ant entirely meta all the time or a bacterium is that like a total meta perspective and they don't have to go through this like ridiculous like self like whatever creating a dissolving loop or danny would know better than i would you actually have to ask the ants that's kind of the hard part but we can make one comment the difference between the egocentric and then the more impartial or autobiographical reminded me a lot of the projective consciousness model projective geometry model which was also integrated with higher order theories and global workspace and a lot of those other topics with um ryan smith and christopher white and also the more geometrical work um with rudraf and others because it's like we do need both those modes we need the one where the hands are bigger when they're closer to your face and we need the one where it's like you're playing the video game with the third person view and then it was really interesting when you point to a lot of the um seen as deficits of memory or cognition like false memory but just seeing them as computational outcomes of stochastic interoceptive lossy compression and generalization semantic encoding what's more likely who kind of person are you that's gonna you know scale whether you think you did something or not so that's just such a interesting area that you kind of took it down thank you it's interesting bringing up like rudolph cause like there's some things like i wonder like you know when do you first get the capacity for projected geometric modeling of what kinds and so like this so with and what's the nature of the projective geometric model like is this something that should i still i can stop the sharing there we go um is this something that you would expect at the level of like cortical micro circuitry like some interpretation of like what cortical macro columns are doing or is this something that's actually more complex and it's like a distributed algorithm like hinton has like ideas like glom that kind of get it something like this like the because what you have with this projective geometric model is the ability to flexibly move between egocentric and allocentric perspectives i can take your point of view or mine once i um raise everything or boost it up into this um i forget that is it hyperbolic or if it's whatever this the space is that he puts it in from within there you can embed either either one and move between them um the thing i'm wondering is like the nature of these like you know i just mentioned this uh like ability to take fictitious points of view on you i'm wondering whether like the way in which we do it is a lot more derived and a kind of skill of moving between things like the representations you might have acquired through mirroring with others and it's and like to to take another perspective rather than it being something that that the visual system can just do of moving between of automatically translating from points of view within this higher space like what is the nature of that model like i've even wondered like for instance is perception actually always 2d and in a kind of like holographic way it's always a 2d projection and when we talk about something like depth it's actually either some sort of like sense of affordance or intercept of cross mapping or the likely transitions between 2d or even like 3d like we like you know we never like mars said it was like 2.5 division like is that just the ability to make sure that what's on the screen that's 2d is always coherent or that you move between as if you can move around a thing but we actually never perceive anything in 3d um in terms of our visual spatial awareness but to loop around to the ant um i don't know but like you might distinguish for instance if you consider the perspective of the ant itself as generative model that might be by default more allocentric especially because of its like what it's specifically doing to the you know it takes something even makes sense to pick it it's it's fundamentally part of the super organism the organism is it's not even super organism like like what like daniel says like it's it is the unit that makes sense it's being selected is the shared enactment but then if you go maybe like the attractor dynamics of like the central complex it's like quasi hippocampal quasi-thalamic thing that might be more egocentric and i don't think you're going to get just totally agree with that i think it's an established fact that um the compass heading or the polarization of light is processed within one nest mate head and then there are collective decisions like the sex ratio of the larva that they're going to keep or the overall regulation of foraging that wouldn't be expected to have even a correlate necessarily in the head capsule of any single nest mate and so i think that's where the discussion comes to like multi-scale systems which is kind of a fun topic because in multi-scale systems we see sometimes a clean mechanistic difference between the lateral and the kind of vertical or the nesting of systems so like communication we can see the interface where you have the lateral layer of individuals and then where you have the layers that are inside of the individual it's a little harder to disentangle it if they're all like kind of wire but what are you or blue do you want to ask a question otherwise yeah i mean it just makes me think of how like are we fundamentally broken with our like self little egocentric loop right like are we maybe like measuring forever the incorrect scale of stuff like should we maybe be more concerned about like the scale of all of us together or the scale of all of the parts that make up us instead we like kind of falsify and create like our ego and our relationship we like project a lot about how we think we should be within the world or how we think we things are within us that like we don't actually know anything about i don't know just related to that i mean it's definitely beyond neuroscience but like i'd say uh yes but also um like i both think this is one of the most valuable and fraught parts of uh psychedelic and discourse around psychedelics and meditation also where it's like if you get this myopic egocentric perspective that's gonna be a real problem um for multiple reasons it's even for you it's gonna be a problem because you see things from one way you're gonna get owned by yourself as a kind of like you'll get trapped in all sorts of local you see things in one way you really barely see it anyway um i do worry a little bit though about um overly you know egos get themselves in trouble but like the ability to stitch together for instance you know there is like a unit over which you have potentially maximum purchase and then and like in terms of influence there's a sense in which like you're the steward of that because like the causal closure is densest and there's also a sense in which like let's say um i want to coordinate with others including myself across time like the ability to like have a narrative structure of the ability that's going to be helpful for not like just jumping at whatever source of utility or or expected free energy comes along and so like to basically avoid being um functionally psychopathic like you kind of need um the ego but you also need to have the ego not be too tight to avoid being functionally psychopathic so it's like not too tight not too loose otherwise the whole thing's wrecked that sounds like narrative stigmergy like agent environment feedback loops also of the script research that we've talked about previously with alder russen and others where that's sort of the top down prior so it's related to what you said about the hard problem as to if or why or why not there would be awareness of experience at for example the bodily level versus the you know tripartite zoom chat level um but at the very least instrumentally those top level scripts are the narratives that decide like are we going to put everybody you know through a a tunnel where only this exact width is going to make it through or is that a phenotype that we have a lot ambiguity around using the bayesian prior term so um it is interesting how and i almost want to ask you whether you started with the more neuroside and found it broaching to these questions or whether you started maybe focus on other areas and then saw the brain's specifics as increasingly important like there's so little separation for me between like the neuroscience obsession and the existential like i'm constantly doing i'm constantly in a state of existential freakout and neuroscience obsession that i don't know which led which i don't know who's the dog who's the tail and who's wagging whom it's just constant existential crisis and obsession wait so i don't believe what would be advice to somebody who's curious about the field or they want to be working in an interesting area of neuroscience and embodiment and everything that you kind of mentioned today it seems like if someone wants to focus on psychedelics the best route is to uh cut your teeth or to like basically get a home outside of psychedelics establish a competency there and then try to bring that within the psychedelic domain that seems like the best path most people who succeed in the space take yeah so it's and you know the standard fraud thing of academia where it's like you're trying to not be an inch wide and a mile deep and not be a jack of all trades master of none and where the perverse incentives abound and good luck um uh like some combination of like both paying your dues and not optimizing yourself into a brittle straight jacket where you can't see anything past the next grant yikes i feel you thanks for the answer so one small question from the chat and then if anyone else has any final questions so adam thanks for your time and the great presentation and cool work and hanging out with us so adam mentioned the dmt paper with respect to the cortical traveling waves the question was wonder if there's any other work on non-alpha waves and then just if you know about that and then just a little more generally why do these traveling waves matter or what matters if they are important or not like i drew like those swaths those like synchronous complexes but those are necessarily a core screening you zoom in on them and it's like so what's the ontological status it has a certain relational realism and that the extent that makes sense like draw that out it's active inference in general with markov blankets it's like you can it's relative to what other system that's engaging in what types of inference they'll just decide like the scope of relevance i mean there's additional like autopoetic self-organizing elements of like in the internal but uh so if you look though with like a synchronous like an alpha like beta alpha you zoom in it's actually comprised of these traveling waves as more of a fine-grained description although you zoom in on that that's actually corresponding of population level activity um and so it'll depend so there's going to be potentially if you're looking at traveling in cortical waves that might be useful so i focus on like the standing wave description as like um a joint belief over whatever is in its scope but if you're talking about something like predictive processing within the model you you might want to zoom in on this standing wave or the standing of description for like the business end of passing on your predictions and then kicking up your prediction errors but depends like it depends on the the the within the multiscale account what's of interest for the modeler one quick thing from this in the side uh handle the alpha wave question whether there's any non-alpha wave traveling wave papers i don't think so i think the one i think the terminal one was focusing on alpha if i'm remembering i could be misremembering i think but there are four psychedelics um uh i terry sanowski i think would be the person to look at for some really good general biophysics of traveling waves in the brain yeah sanowsky i think would be the place to look a quick aside on on dmt like i mentioned like there might be endogenous cmt um there's debate over how significant it is so some people say it's not significant at all some people say it's very significant so it's one place we know that it's produced is in the pineal body the seed of the soul and it's another place and there's some evidence but some people have questioned it that it might be um almost akin to like a neurotransmitter system in its own right we just didn't know how to look at it before um jimoh borgen at university of michigan if you're interested in that issue and that debate uh but it's possible though like um one of the main endogenous ligands we don't know of the five hd2a system one of the main stimulus might not actually be serotonin but might be dmt so like under like let's say like a limit experience like uh you know so one like i mentioned like carbon dioxide can do it or maybe lactic acid like you're you're running for your life or for your lunch and it's super high levels of blood acidity and that might be activating these proto-psychedelic pathways um but another one could be like potentially dmt mediated and things like um sex earthing and near-death experiences so like you almost bled out um maybe it's time to relax your beliefs about to bond with another creature for years maybe this is how part of how you do it um whether bonding with a mate or your offspring very expected these are early days interesting and also made me think about brain and body dopamine and serotonin other neurotransmitters in the blood in the blood vessels the immune system and then neuropeptides the whole continuum of short acting neurotransmitters to slower acting phenomena and then oh we're thinking about it from a nested multi-scale systems perspective with this kind of more local faster and then slower broader spatial scales which just has a lot of analogy with everything from regional governance to the regulation of tissue so it's kind of an interesting idea how a lot comes together there it also made me think of how staring into someone else's eyes for like a prolonged period of time induces like violent disturbing hallucinations which is like reported in i don't know some paper i can't remember but i'll i'll look it up and and see if i can post it in the chat but i'm really interested how like thinking through others minds or like looking through others eyes um is is so violently incorrect to us that it's perceived as like an altered perception i i wonder if this is speaking to like the way in which we bootstrap um like objectified selfhood and and me and like but but because like we do it so much this is like a core axis of imagination and perception that you can create like these weird feedback loops and i also wonder like how much it varies across people like like if you're like really far on the autism spectrum like or or far in a schizophrenia spectrum like would you like in a moment like you look in someone's eyes and it's like it takes you two seconds to enter like the psychedelic space or might take like you know you got to sit there for like an hour okay this test was like i think 10 minutes so i'll i'll find it right now post it in the chat it's happening to me right now well adam not your first active livestream not your last so thanks for joining and sharing this cool work we hope to see you around the lab and just in the area so always a pleasure and an honor talk to you later